AI for Good Sandbox Hackathon — ARGOS, container tracking and safety

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1. Introduction

A container passes through several stages, locations, and sometimes different modes of transport. When its position, condition, or an alert arrives separately, the person tracking the shipment may lack the context needed to respond. ARGOS brings these details together around an IoT unit attached to the container and a tracking application. The project is intended to make movements and observed events during transport visible in a tool suited to operations managed from Central Africa.

For the AI for Good case, the demonstration is organized around containerized cargo moving from Douala to a fictional inland yard in Cameroon. Its core is a coordination of four simulated AI roles—operator, customs-port, data protection, and adviser—working from staged facts and mandate-specific sources. The ARGOS unit is an optional evidence channel; it is not presented as a decision-maker or public authority.

ARGOS combines an ESP32 device, sensors, an ingestion server, and a web application. The application presents GPS positions, history, the planned route, sensor readings, and notifications with their trigger. It can also show an estimated arrival time based on the planned route and recent GPS speed. This ETA is a deterministic projection, not a learned prediction. The logistic classifier currently trained is experimental, uses synthetic scenarios, and does not power operational alerts. Software agents and the LLM help bring together and explain authorized information; they do not replace safety rules or an operator's judgement.

2. Description of the use case

An ARGOS unit transmits positions, speed, and readings such as temperature, humidity, shock, door opening, flame detection, and battery level. The server checks the message structure, timestamp, sequence, and its association with a box. It stores the readings and positions, then presents them in the application. Configurable rules flag values or events that cross set thresholds. Route tracking compares positions with a planned corridor; configured zones and port reference points provide indicative geographic context.

The interface keeps notifications brief. It states what triggered the alert and leaves the readings, time, position, and full context to the detail view. For arrival time, ARGOS projects the remaining travel time by dividing the distance along the planned route by the median of recent GPS speeds. The calculation is suspended when the route or positions are missing, the latest position is too old, or the unit has stopped. It does not model traffic, weather, network outages, borders, customs or port queues, or transshipments. The screen labels the result as an indicative estimate and shows its freshness; it is not a guaranteed arrival time.

The registered agents cover IoT readings, GPS and routes, deterministic signal assessment, document search, transport regulatory references, and port context. When Gemini is configured, the LLM can select tools and write a summary from their results. The provider's availability and use with real data still need validation for the deployment. Rules and data ingestion continue to work independently of the LLM's wording.

ARGOS addresses the lack of context between a reading, a position, a route, and the event shown to an operator. The project does not claim to replace carrier platforms, terminal data, or shipment managers' decisions. It provides tracking from an instrumented box and distinguishes observed facts, rule-based calculations, and the limits of estimates.

3. Use case requirements

- SRC — Sources: the unit sends available readings with timestamps, sequence numbers, and quality indicators. The representativeness of real readings, sensor calibration, and mounting conditions still need to be established in the field.
- C — Collector: the API collection point receives validated ESP32 messages and passes them to pre-processing. Wi-Fi or a local connection transports those messages; neither defines Y.3172 node C. Synchronization after an outage still needs to be demonstrated.
- PP — Pre-processing: the server validates fields and ranges, binds readings to the device and box, rejects repeated sequences, and neutralizes some values the firmware marks invalid. Synthetic data is identified as such and must not be presented as field observation.
- M — Methods: threshold and risk alerts use deterministic rules; route tracking uses geometry and GPS points; ETA uses route distance and recent speeds. A binary logistic regression was trained offline on 240 synthetic samples from four scenarios. The artifact is not loaded by the application and is not a field-validated predictive model.
- P — Protection and decisions: the API checks the user profile, company, and authorized containers; operational tools are limited to selected boxes. A person remains responsible for decisions. Retention, transfers, contractual responsibilities, and secondary use of data still require organizational and legal review.
- D — Deployment: the application uses Flask, PostgreSQL/TimescaleDB, and Redis, with containerized processes. The demonstration does not prove availability, disaster recovery, or network coverage for a regional deployment.
- SINK — Interface: the map and detail views show the latest point, available history, route, indicative ETA, and full notification details. The operator checks the information and chooses what to do next.

4. Mapping to ITU-T Y.3172, AI Readiness 2.0, and sector references

Node	ARGOS use case	AI Readiness 2.0 / relevant dimension	Documents, policies, and standards to consider
SRC	ESP32 unit attached to the container; GPS and temperature, humidity, shock, door, flame, and battery sensors. Quality fields and data origin accompany readings when transmitted.	Data sources, accessibility, quality, representativeness, and labelling; availability of devices and sensors.	African Union Continental AI Strategy for capacity, data, and regional cooperation; sensor calibration protocols; equipment register, provenance, consent, and data-use agreements.
C	API-side event collector: receives validated IoT messages and passes them into the ARGOS pipeline. Wi-Fi and direct local links are input transports, not the node name.	Collection availability, underlying connectivity, devices, edge/cloud, and digital infrastructure.	API contract, device checks, collection log; outage, recovery, and synchronization exercises.
PP	Schema validation, timestamp and sequence checks, box/container binding, quality normalization, and PostgreSQL storage.	Data quality, harmonization, interoperability, governance, and protection.	ARGOS data dictionary and schema version; quality and provenance rules; country-by-country review of Cameroon Law No. 2024/017 and rules applicable to other corridor jurisdictions.
M	Deterministic threshold and risk rules; geometric route calculations; ETA from distance and recent speed; six specialist capabilities; configurable LLM; experimental logistics classifier trained only on synthetic data.	Data and model provenance, metadata, evaluation, contextualization, and risk management.	NIST AI RMF as a voluntary reference; evaluation protocol by shipment and corridor; model card, versioning, field metrics, and usage limits. The synthetic classifier is not a production model.
P	Server-side checks restrict access by profile, company, and authorized box. Alerts show their triggers; the operator reviews the situation. LLM outputs may be sent to the configured external provider.	Trust, accountability, human oversight, privacy, and provider governance.	Role-based access policy; review and contestation process; retention and deletion rules; provider contracts and transfer review; legal review for each jurisdiction.
D	Flask API and processing services; PostgreSQL/TimescaleDB storage; Redis and a worker for application tasks. The demonstration configuration does not establish reliable regional operation.	Deployment, computing resources, availability, energy consumption, and operations.	Hosting plan, backup and restore, incident response, business continuity, energy use, and costs measured on the pilot corridor.
SINK	Web application with a Leaflet map, container list, routes, positions, indicative ETA, brief notifications, and detail views.	Human interface, local languages, accessibility, and collaboration between people and AI.	Operator guide, accessible mobile design, French/English as needed by teams, and a procedure for checking alerts and correcting data.

This mapping follows the pipeline structure of the supplied template and connects each node to relevant readiness dimensions. The two ITU references are different frameworks: ITU-T Y.3172 describes an ML management architecture for future networks, including IMT-2020; the AI Ready report analyses readiness across data, infrastructure, skills, policy, and innovation ecosystems. ARGOS is neither certified against Y.3172 nor assessed with an AI Readiness score. The table locates existing evidence and work still to be done.

Policy references do not amount to a compliance claim. The African Union strategy provides context for AI adapted to African priorities and capabilities. Cameroon's personal data law needs to be reviewed against the facts and roles of the processing; it does not determine the rules for every Central African corridor. SOLAS/VGM and the CTU Code are separate transport references: ARGOS does not weigh containers, certify their securing, or determine whether those rules apply.

5. Error and controversy scenarios

Apparent route deviation. A GPS position appears to leave the corridor. ARGOS shows the point, accuracy, time, and geometric score in the details. The operator checks GPS quality and the planned route before contacting the carrier. The score is not a probability of theft or an incident.

Disputed sensor alert. A shock or opening triggers a notification even though the sensor is loose, invalid, or affected by a brief spike. The details show the reading and threshold used; the responsible person checks the sensor and context. The alert alone does not establish damage or fault.

Network outage and stale ETA. The ESP32 stops sending positions in an area without coverage. ARGOS shows the age of the last point and suspends the estimate when data is too old. It must not extrapolate a position as if it had just been measured. Offline storage and synchronization need field testing before anyone relies on them.

Port proximity mistaken for official entry. A GPS point approaches a port reference point. ARGOS may show indicative proximity; it does not present this as official entry. If a documented geofence is configured, its boundary source and GPS accuracy still need to be checked.

Using positions to sanction a driver. A company wants to turn locations and alerts into an individual ranking or automatic sanction. Location data may concern identifiable people. The purpose, notice, access, retention period, and applicable legal basis must be established before such use; an ARGOS alert alone is not grounds for a sanction.

Operational data sent to an LLM. An assistance request may send telemetry or position results to the configured Gemini provider. Before enabling this for real data, the organization must review what is sent, the contract, transfers, retention, and minimization measures. Until those points are reviewed, assistance should be limited to demonstration data or disabled for those data.

In these scenarios, readings, freshness, and provenance remain available for review. An operator can request a check and report an incorrect reading. The project does not yet demonstrate a complete process for contestation, erasure, or field investigation; those responsibilities must be assigned before deployment with customers.

Staged scenario silent_container_v1. Six synthetic steps introduce a declaration and seal, terminal exit, a door/temperature alert and uncertain position, a location-sharing request, a delayed yard reading, and a proposed human inspection. A1, A2, and A3 receive different facts; A4 reconciles their summaries against permitted passages. At T3, location sharing remains blocked until a person reviews purpose, claimed basis, recipients, retention, and safeguards. A trace keeps event IDs, source passages and pages, uncertainty, recommendations, and review status. The scenario can be replayed offline with scripts/run_policy_scenario.py.

6. Policy and standards gaps derived from the mapping exercise

The ARGOS source register contains a bounded summary of selected WTO Trade Facilitation Agreement provisions and visually checked excerpts of Articles 9–16, 19, 21, 22, and 25 of Cameroon Law No. 2024/017, transcribed from an official scanned copy. These excerpts support questions about consent and exceptions, purpose, retention, notices, transfers, automated decisions, and data breaches. They are not the full law and have not received an independent legal review. Applicability, implementing rules, the status of the authority, and customs responsibility remain unresolved whenever the relevant source has not been retrieved.

The ARGOS classifier is binary logistic regression trained on four synthetic scenarios: 240 points in total, 173 stable and 67 simulated anomalies. In leave-one-scenario-out validation, reported mean recall is 56.3% and mean precision is 100% against the simulated labels only. This measures agreement with the simulation, not the ability to detect real incidents. The 61-point Central Africa replay is synthetic as well; the streaming script records it in ARGOS but does not retrain the classifier. The application does not use the model artifact for its alerts. No operational ARGOS model has been validated to predict incidents or delays.

The current ETA is a transparent calculation, but it has not been compared with actual arrival times. Border and port queues, weather, traffic, transshipments, route stages, and network outages are not modelled. Synthetic scenarios do not measure sensor accuracy during transport or representativeness across Central African routes, seasons, devices, and operators.

The next technical step is to collect real trips with provenance, quality readings, actual departure and arrival times, and incidents confirmed by operators. Each target must be defined separately: incident detection, delay prediction, and arrival-time estimation are different modelling tasks. Evaluation should hold out complete shipments and compare each model with the current rules or ETA. Without those observations, a claim of regional predictive performance would have no basis.

The review of personal data and transfers to providers is not complete. The code checks some access boundaries, but it cannot by itself establish purpose, legal basis, retention periods, notices, or contracts. The regulatory catalogue is a local document search and does not automatically determine whether a text applies to a shipment. Cameroon's rules, those of transit countries, and contractual commitments must be reviewed for each deployment.

International commercial platforms, including project44, describe visibility built from carrier integrations, port events, and vessel/AIS data, with network-based ETAs and alerts. ARGOS focuses on telemetry from a box attached to the container, sensor condition, and company-scoped access. The repository contains no independent benchmark comparing accuracy, coverage, or cost. This difference in scope does not prove ARGOS is superior; it shows the external integrations needed to cover an end-to-end journey.

References used in this mapping include the African Union Continental AI Strategy (https://au.int/en/documents/20240809/continental-artificial-intelligence-strategy), the ITU AI Ready report (https://www.itu.int/epublications/publication/ai-ready-analysis-towards-a-standardized-readiness-framework), ITU-T Y.3172 (https://www.itu.int/rec/T-REC-Y.3172/en), the NIST AI Risk Management Framework (https://www.nist.gov/itl/ai-risk-management-framework), Cameroon Law No. 2024/017 (https://www.prc.cm/files/2b/9f/21/1055fa3c2251b4c4248fd301f584daaf.pdf), the WTO Trade Facilitation Agreement (https://www.wto.org/English/docs_e/legal_e/tfa-nov14_e.htm), IMO references on SOLAS/VGM (https://www.imo.org/en/ourwork/safety/pages/verification-of-the-gross-mass.aspx) and the CTU Code (https://www.imo.org/en/ourwork/safety/pages/ctu-code.aspx), and project44's public description of ocean visibility (https://www.project44.com/platform/visibility/ocean/). These documents provide context; their inclusion does not certify ARGOS.